

① Estimate parameters $\theta = [\theta_1, \theta_2]$ for a random variable $X \in \{1, 2, 3\}$ with $P(X=1) = \theta_1$, $P(X=2) = 2\theta_1$, $P(X=3) = \theta_2$ and counts s_1, s_2, s_3 from data D .

② Identify the valid probability constraints on θ_1, θ_2

Probabilities must be non-negative and sum to 1.

Non negativity: ~~θ_1~~ $\theta_1 \geq 0$, $\theta_2 \geq 0$

Normalization: $\theta_1 + 2\theta_1 + \theta_2 = 1 \Rightarrow 3\theta_1 + \theta_2 = 1$

Equivalently $\theta_2 = 1 - 3\theta_1$ with $0 \leq \theta_1 \leq \frac{1}{3}$

③ Joint Probability and log-likelihood

Joint Probability of the specific ordered sequence

$$P(D|\theta) = \prod_{i=1}^n (P(x^i|\theta)) = \theta_1^{s_1} (2\theta_1)^{s_2} \theta_2^{s_3}$$

log-likelihood

$$\begin{aligned} \log P(D|\theta) &= s_1 \log \theta_1 + s_2 \log (2\theta_1) + s_3 \log \theta_2 \\ &= (s_1 + s_2) \log \theta_1 + s_2 \log 2 + s_3 \log \theta_2 \end{aligned}$$

④ Maximum likelihood Estimation $\hat{\theta}$ of θ

$$\ell(\theta_1, \theta_2) = (s_1 + s_2) \log \theta_1 + s_2 \log 2 + s_3 \log \theta_2$$

Use $\theta_2 = 1 - 3\theta_1$, to get a 1-D problem

$$g(\theta_1) = (s_1 + s_2) \log \theta_1 + s_3 \log (1 - 3\theta_1) + \text{const.}$$

set derivative to zero:

$$\frac{s_1 + s_2}{\theta_1} - \frac{3s_3}{1-3\theta_1} = 0$$

$$\Rightarrow (s_1 + s_2)(1 - 3\theta_1) = 3s_3\theta_1$$

$$\Rightarrow \hat{\theta}_1 = \frac{s_1 + s_2}{3n}$$

$$\text{Then } \hat{\theta}_2 = 1 - 3\hat{\theta}_1 = 1 - \frac{s_1 + s_2}{n} = \frac{s_3}{n}$$

$$\hat{\theta}_1 = \frac{s_1 + s_2}{3n} \quad \hat{\theta}_2 = \frac{s_3}{n}$$

② Given iid $x^{(1)}, \dots, x^{(n)} \sim f(x|\beta) = \frac{1}{\beta} \exp(-x/\beta)$
for $x \geq 0$

$$\text{let } S = \sum_{i=1}^n x^{(i)}$$

Likelihood

$$L(\beta) = \prod_{i=1}^n \frac{1}{\beta} \exp\left(-\frac{x^i}{\beta}\right) = \frac{1}{\beta^n} \exp\left(-\frac{S}{\beta}\right)$$

log-likelihood

$$l(\beta) = -n \log \beta - \frac{S}{\beta}$$

Differentiate and set to zero

$$\frac{dl}{d\beta} = -\frac{n}{\beta} + \frac{S}{\beta^2} = 0 \Rightarrow \beta = \frac{S}{n} = \bar{x}$$

Second derivative:

$$\frac{d^2 l}{d\beta^2} = \frac{n}{\beta^2} - \frac{2S}{\beta^3}$$

$$\text{at } \beta = \hat{\beta} = \frac{S}{n}, \text{ this equals } \frac{-n^3}{S^2} < 0$$

which tell it is a maximum as it is less than zero.

$$\hat{\beta} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x^i$$

③

(a) Identifying the posterior for a defect rate with a uniform prior

Model: $x \sim \text{Binomial}(n=30, \theta)$ with observed $k=5$ defective

Prior: $\theta \sim \text{Uniform}(0, 1)$.

Likelihood (up to a constant not depending on θ)

$$L(\theta) \propto \theta^k (1-\theta)^{n-k} = \theta^5 (1-\theta)^{25}$$

Posterior

$$P(\theta|D) \propto \underbrace{\theta^5 (1-\theta)^{25}}_{\text{likelihood}} \underbrace{\theta^{1-1} (1-\theta)^{1-1}}_{\text{uniform prior}} = \theta^5 (1-\theta)^{25}, 0 \leq \theta \leq 1$$

⑥ Calculate Posterior $p(\mu|D)$ for $x^i \sim N(\mu, 1)$ with prior $\mu \sim N(0, 1)$

Let $D = \{x^{(1)}, \dots, x^{(n)}\}$, $S = \sum_{i=1}^n x^{(i)}$ and $\bar{x} = S/n$

Posterior Variance

$$V_n = \frac{1}{\text{Prior precision} + n \cdot \text{likelihood precision}}$$

As likelihood precision per datum is 1. Prior Precision is 1

$$V_n = \frac{1}{1+n}$$

Posterior mean:

$$m_n = V_n (1 \cdot 0 + 1 \cdot S) = \frac{S}{n+1} \Rightarrow \frac{n}{n+1} \bar{x}$$

Therefore

$$\mu|D \sim N\left(\frac{n}{n+1}\bar{x}, \frac{1}{n+1}\right)$$